Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 38: Graphs as Data

Adjacency Matrices and Families of Graphs

Chartrand & Zhang, Section 12.1

Name: _____ Date: _____

Where we are

A picture of a graph is convenient for people and useless to a computer. Today: the standard way to store a graph as an array of numbers — and a surprising fact about what happens when you multiply that array by itself.

By the end of today you should be able to

1. Write the adjacency matrix of a graph and read a graph off a matrix.
2. Use A^k to count walks.
3. Recognise K_n, C_n, P_n, and bipartite graphs.

1. The adjacency matrix

Adjacency matrix

For a graph G of order n with $V(G) = \{v_1, \ldots, v_n\}$, the **adjacency matrix** is the $n \times n$ matrix $A = [a_{ij}]$ with

$$a_{ij} = \begin{cases} 1 & \text{if } v_i v_j \in E(G) \\ 0 & \text{otherwise.} \end{cases}$$

Three properties, each with a reason.

Property	Why
The main diagonal is all 0	
A is symmetric	
Row i sums to $\deg v_i$	

Build one. $V = \{v_1, \ldots, v_4\}$, $E = \{v_1v_2, v_1v_3, v_2v_3, v_3v_4\}$.

	v_1	v_2	v_3	v_4
v_1				
v_2				
v_3				
v_4				

Total of all entries: _____ $= 2m$, so the size is _____.

2. Powers of A count walks

Walk

A **walk** of length k from u to v is a sequence of $k+1$ vertices starting at u and ending at v, with consecutive vertices adjacent. Vertices and edges *may* repeat.

Theorem 12.22

For a graph with adjacency matrix A, the number of v_i–v_j walks of length k is the (i, j) entry of A^k.

For the matrix above,

$$A^2 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \qquad A^3 = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 4 & 1 \\ 4 & 4 & 2 & 3 \\ 1 & 1 & 3 & 0 \end{bmatrix}$$

Question	Answer
Walks of length 2 from v_1 to v_4	
Why is $a_{33}^{(2)} = 3$?	
Walks of length 3 from v_2 to v_3	

The diagonal of A^2 is the degree sequence

A walk of length 2 from v back to v goes out along an edge and straight back, so there is one for each edge at v. That is why $a_{ii}^{(2)} = \deg v_i$ — and it gives a second way to read degrees off the matrix.

3. Families worth naming

Name	Description	Order and size
K_n	complete: every pair adjacent	
C_n	cycle on n vertices	
P_n	path on n vertices	
$K_{s,t}$	complete bipartite	

K_n is $(n-1)$-regular, and its size $\binom{n}{2}$ is the handshake count from Week 13 — the same number, because an edge *is* an unordered pair.

Bipartite

G is **bipartite** if $V(G)$ can be split into two sets so that every edge joins a vertex in one to a vertex in the other — no edge inside either set.

A theorem worth knowing: a nontrivial graph is bipartite **if and only if** it contains no _____.

So C_4 is bipartite and C_5 is _____. Checking for an odd cycle is far easier than searching for a split.

Exit ticket

1. What is the size of K_7? Of $K_{3,4}$?

2. If A is an adjacency matrix, what does the $(2,2)$ entry of A^2 tell you?

Before Friday

Read Section 12.2, "Connected Graphs," pp. 444–455.

Self-check (not collected). Section 12.1, Exercises 9, 11, and 13 — odd-numbered, answers in the back.

Homework 13 is due Friday, November 20, 11:59 p.m.

Questions I still have